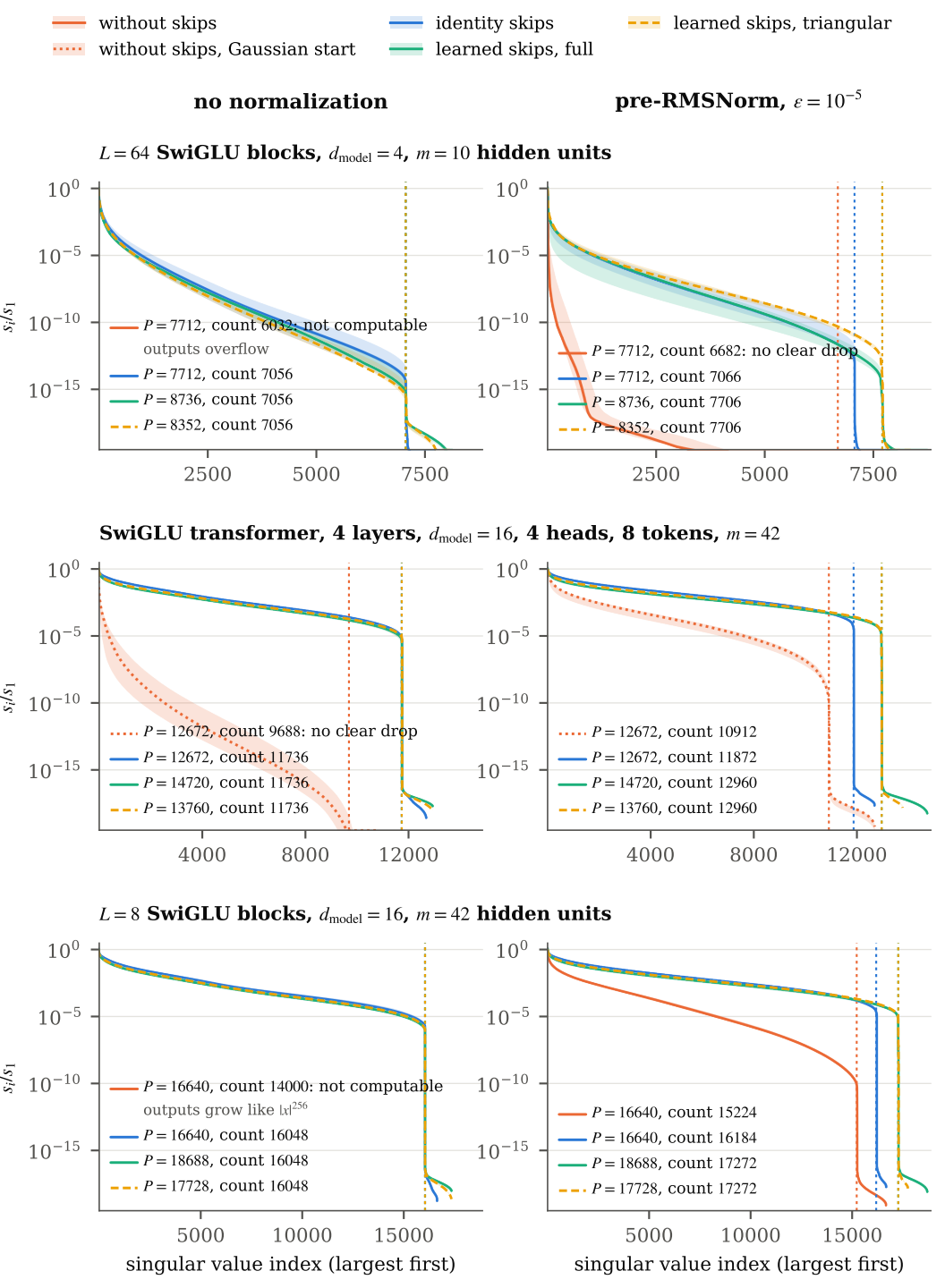

without skips
without skips, Gaussian start
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ SwiGLU blocks, $d_{\text{model}} = 4$, $m = 10$ hidden units
s_i/s_1
$P = 7712$, count 6032: not computable
outputs overflow
$P = 7712$, count 7056
$P = 8736$, count 7056
$P = 8352$, count 7056
$P = 7712$, count 6682: no clear drop
$P = 7712$, count 7066
$P = 8736$, count 7706
$P = 8352$, count 7706
SwiGLU transformer, 4 layers, $d_{\text{model}} = 16$, 4 heads, 8 tokens, $m = 42$
$P = 12672$, count 9688: no clear drop
$P = 12672$, count 11736
$P = 14720$, count 11736
$P = 13760$, count 11736
$P = 12672$, count 10912
$P = 12672$, count 11872
$P = 14720$, count 12960
$P = 13760$, count 12960
$L = 8$ SwiGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$P = 16640$, count 14000: not computable
outputs grow like $|x|^{256}$
$P = 16640$, count 16048
$P = 18688$, count 16048
$P = 17728$, count 16048
$P = 16640$, count 15224
$P = 16640$, count 16184
$P = 18688$, count 17272
$P = 17728$, count 17272
singular value index (largest first)
singular value index (largest first)